Enstrophy and Weighted-Enstrophy Balance Identities on $\mathbb{T}^3$: a Lean 4 Formalization

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Abstract

We formalize in Lean 4 with Mathlib two exact balance identities from the a-posteriori Lyapunov functional approach to the 3-D incompressible Navier–Stokes vorticity equation on the torus: the enstrophy identity $\frac{dE}{dt} = P - \nu \int |\nabla\omega|^2$ and the weighted identity for $Q = \int |\omega|^4\phi$. Both are proved for every periodic differential algebra whose integral annihilates spatial derivatives and commutes with the time derivative, and unconditionally for trigonometric polynomials on $\mathbb{T}^3$. The development has 0 `sorry` and uses only the axioms `propext`, `Classical.choice`, `Quot.sound`.

1 Introduction

Commentary (adapted from the author's paper). Classical energy methods for the Navier–Stokes equations control the kinetic energy but not the enstrophy in three dimensions; the vortex-stretching term can produce growth that is difficult to absorb. The approach studies the exact classical equations together with an auxiliary scalar field ϕ transported by $|\omega|^2$, and the two integrals $E = \frac{1}{2}\int |\omega|^2$ and $Q = \int |\omega|^4\phi$. The first step is to compute dE/dt and dQ/dt exactly. Those two computations are what this paper formalizes. They are identities: no inequality and no bound on the solution is involved.

2 Setting

Let $u, \omega : \mathbb{T}^3 \to \mathbb{R}^3$ and $\phi : \mathbb{T}^3 \to \mathbb{R}$ depend on t, with

$$\partial_k u_k = 0, \tag{1}$$
$$\partial_t \omega_i = -u_k \partial_k \omega_i + \omega_k \partial_k u_i + \nu\, \partial_k \partial_k \omega_i, \tag{2}$$
$$\partial_t \phi = -u_k \partial_k \phi + |\omega|^2, \tag{3}$$

summing over repeated indices. The relation $\omega = \nabla \times u$ is not needed for either identity.

Definition 1 (Periodic setting, `PeriodicSetting`)**.** A commutative K-algebra A with K-derivations $\partial_1, \partial_2, \partial_3, \partial_t$, a K-module B, a K-linear map $I : A \to B$ and a K-linear $D_t : B \to B$ such that $I(\partial_k f) = 0$ for all k, f and $I(\partial_t f) = D_t(If)$.

For smooth periodic fields, $I = \int_{\mathbb{T}^3}$, $D_t = \frac{d}{dt}$; the two axioms are the divergence theorem on the torus and differentiation under the integral.

Definition 2 (Trigonometric model, `Trig.setting`)**.** $A = R[\mathbb{Z}^3]$ (finite sums $\sum_n f_n e^{2\pi i n\cdot x}$, $f_n \in R$), where R is a commutative $\mathbb{C}$-algebra with a derivation D. Set $\partial_k(\sum f_n e_n) = \sum 2\pi i n_k f_n e_n$, $\partial_t(\sum f_n e_n) = \sum D(f_n)e_n$, $I(f) = f_0$, $D_t = D$.

Lemma 3 (`dx`, `dt`)**.** *∂_k and ∂_t are derivations of $R[\mathbb{Z}^3]$.*

Proof. Both are linear, so the Leibniz rule reduces to monomials. For ∂_k: $\partial_k(ae_n \cdot be_m) = 2\pi i(n_k + m_k)ab\, e_{n+m} = ae_n \cdot 2\pi i m_k b e_m + be_m \cdot 2\pi i n_k a e_n$. For ∂_t: $D(ab)e_{n+m} = (aDb + bDa)e_{n+m}$. □

Lemma 4 (Torus divergence lemma, `integral_dx`, `integral_dt`)**.** *$I(\partial_k f) = 2\pi i \cdot 0 \cdot f_0 = 0$ and $I(\partial_t f) = D(f_0) = D(If)$.*

Hence the trigonometric model is a periodic setting, and every theorem below holds for it.

3 Pointwise lemmas

Lemma 5 (`adv_eq`)**.** *$u_k\partial_k g = \partial_k(u_k g) - (\partial_k u_k)g$. Hence (`int_adv_eq_zero`) if (1) holds, $I(u_k\partial_k g) = 0$.*

Proof. Leibniz: $\partial_k(u_k g) = u_k\partial_k g + g\partial_k u_k$. Apply I and use $I \circ \partial_k = 0$ and (1). □

Lemma 6 (`wLapW_eq`)**.** *$\omega_i\partial_k\partial_k\omega_i = \partial_k(\omega_i\partial_k\omega_i) - \partial_k\omega_i\,\partial_k\omega_i$.*

Proof. Leibniz on $\partial_k(\omega_i \cdot \partial_k\omega_i)$. □

Lemma 7 (`two_w_adv`)**.** *$2\,\omega_i\,u_k\partial_k\omega_i = u_k\partial_k|\omega|^2$.*

Proof. $\partial_k(\omega_i\omega_i) = 2\omega_i\partial_k\omega_i$. □

Lemma 8 (`stretch_eq`)**.** *$\omega_i\omega_j\partial_j u_i = \omega \cdot (\omega \cdot \nabla)u$.*

4 Identity (1)

Lemma 9 (`dt_half_sq`, `dt_vsq_div`)**.** *Under (1), (2):*

$$\partial_t|\omega|^2 = 2\,\omega_i\omega_j\partial_j u_i - 2\nu\,|\nabla\omega|^2 + \partial_k\big(2\nu\,\omega_i\partial_k\omega_i - u_k|\omega|^2\big).$$

Proof. $\partial_t|\omega|^2 = 2\omega_i\partial_t\omega_i$. Substitute (2): $2\omega_i\partial_t\omega_i = -2\omega_i u_k\partial_k\omega_i + 2\omega_i\omega_k\partial_k u_i + 2\nu\,\omega_i\partial_k\partial_k\omega_i$. By `two_w_adv` and `adv_eq` with (1), the first term is $-\partial_k(u_k|\omega|^2)$. The second is $2\omega_i\omega_j\partial_j u_i$. By `wLapW_eq`, the third is $2\nu\,\partial_k(\omega_i\partial_k\omega_i) - 2\nu|\nabla\omega|^2$. □

Commentary. The transport term vanishes by periodicity and $\nabla \cdot u = 0$; the viscous term integrates by parts to $-\nu\int|\nabla\omega|^2$; the stretching term remains.

Theorem 10 (Identity (1), `enstrophy_identity`, `enstrophy_identity_T3`)**.** *With $E = \frac{1}{2}\int|\omega|^2$ and $P = \int\omega_i\omega_j\partial_j u_i$,*

$$\frac{dE}{dt} = P - \nu\int_{\mathbb{T}^3}|\nabla\omega|^2.$$

(Formalized as $D_t\,I(|\omega|^2) = 2\,I(\omega_i\omega_j\partial_j u_i) - 2\nu\,I(|\nabla\omega|^2)$.)

Proof. $D_t I(|\omega|^2) = I(\partial_t|\omega|^2)$; apply I to the lemma; the divergence term has $I = 0$. □

5 Identity (2)

Write $s = |\omega|^2$, so $|\omega|^4\phi = s^2\phi$.

Lemma 11 (`dt_weighted`)**.** *Under (2), (3):*

$$\partial_t(s^2\phi) = 4s\phi\,\omega_i\omega_j\partial_j u_i + 4\nu\, s\phi\,\omega\cdot\Delta\omega + s^3 - u_k\partial_k(s^2\phi).$$

Proof. $\partial_t(s^2\phi) = 2s\phi\,\partial_t s + s^2\partial_t\phi$. From the proof of (1), $\partial_t s = 2\omega_i\omega_j\partial_j u_i + 2\nu\,\omega\cdot\Delta\omega - u_k\partial_k s$, and by (3) $\partial_t\phi = -u_k\partial_k\phi + s$. So

$$\partial_t(s^2\phi) = 4s\phi\,\omega_i\omega_j\partial_j u_i + 4\nu s\phi\,\omega\cdot\Delta\omega - \big(2s\phi\,u_k\partial_k s + s^2 u_k\partial_k\phi\big) + s^3,$$

and $2s\phi\,u_k\partial_k s + s^2 u_k\partial_k\phi = u_k\partial_k(s^2\phi)$ by Leibniz. □

Commentary (adapted from the author's paper). The transport contributions cancel exactly. Using $\omega\cdot(u\cdot\nabla)\omega = \frac{1}{2}(u\cdot\nabla)|\omega|^2$ they combine into the pure divergence $-\int u\cdot\nabla(\phi|\omega|^4)$, which vanishes on $\mathbb{T}^3$.

Lemma 12 (`dt_weighted_div`)**.** *Under (1) as well,* $\partial_t(s^2\phi) = 4s\phi\,\omega_i\omega_j\partial_j u_i + 4\nu s\phi\,\omega\cdot\Delta\omega + s^3 - \partial_k(u_k s^2\phi)$.

Proof. `adv_eq` with $g = s^2\phi$ and (1). □

Theorem 13 (Identity (2), `weighted_identity`, `weighted_identity_T3`)**.** *With* $Q = \int|\omega|^4\phi$,

$$\frac{dQ}{dt} = 4\int|\omega|^2\phi\,\omega\cdot(\omega\cdot\nabla)u + 4\nu\int|\omega|^2\phi\,\omega\cdot\Delta\omega + \int|\omega|^6.$$

Proof. $D_t I(s^2\phi) = I(\partial_t(s^2\phi))$; apply I, the divergence has $I = 0$, and use `stretch_eq`. □

6 Scope

Both identities are proved for every periodic setting and, with the two integral properties proved, for trigonometric polynomials on $\mathbb{T}^3$ with coefficients in any commutative $\mathbb{C}$-algebra with a derivation. Instantiating the abstract theorems for general smooth periodic fields requires the divergence theorem on $\mathbb{T}^3$ and differentiation under the integral for smooth functions, which are not formalized here. The paper states identities only and makes no regularity claim.

Verification

`lake build`: success, 0 `sorry`. `lake env lean AxiomAudit.lean`: only `propext`, `Classical.choice`, `Quot.sound`. Toolchain `leanprover/lean4:v4.30.0-rc1`, Mathlib `v4.30.0-rc1`.